Apex Evaluation Package

SUS, NASA-TLX, UI/UX instruments and study protocol

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Draft. No data has been collected.

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Forms build guide and master TODO

What to build, in what order, question by question. Follow this top to bottom and the study is ready to pilot.

Four forms. The task script links to three of them; the fourth is completed before the script is opened.

#	Form	When	Times submitted
F0	Consent and demographics	before anything	once
F1	Raw NASA-TLX	after tasks 3, 4, 6, 7, 8, 9	**six times**
F2	SUS	after task 9	once
F3	Apex UX	after SUS	once

Master TODO, in order

Before building anything

1. **Email the supervisor.** Two questions: does this study need an ethics committee submission, and is there a standard IST consent template to use instead of the one in `consent-and-briefing.md`. This gates the first session and nothing else unblocks it.
2. ~~Decide the PT SUS question.~~ **CLOSED.** Validated European Portuguese version, Martins et al. (2015). Paper obtained, ten items transcribed verbatim into `instrument-sus.md`. F2-PT is ready to build.
3. **Decide the TLX response scale.** 0-10 rescaled to 0-100, or a form tool with a real slider. Cannot change mid-study.
4. **Decide the arm split.** Unmoderated only, or unmoderated plus 3-4 moderated sessions.

Build

5. Build F0, F1, F2, F3 as specified below.
6. Test-submit every form once yourself. Open the response spreadsheet and check the column headers are usable. Delete the test rows.
7. Generate six prefilled F1 links, one per task, so the participant does not retype the task name. Google Forms: three-dot menu, "Get pre-filled link", set the task field, copy the URL. Paste each into the matching task in the script.
8. Seed the demo project. Confirm `Export board to CSV` exists and has **never** passed QA. Task 8 does nothing without it.

9. Capture the nine reference screenshots from the seeded project.
10. Fill the four placeholder links into `task-script.md` and export it to PDF.
11. Print participant code cards: code, sign-in credentials, GitHub repo URL, PAT.

Pilot

12. Run **one** pilot participant. Not counted, ever.
13. Check the pilot returned six correctly labelled F1 rows. If not, fix F1 before anyone else runs.
14. Time the pilot. If the tasks overrun 60 minutes, cut an optional task, not a marked one.
15. Fix whatever the pilot broke, then freeze everything. No wording changes after this point.

Run

16. Recruit and run. Reset the demo project between participants and verify the reset.
17. Log any technical failure against the specific task it happened in.

After

18. Export all four response sheets to CSV.
19. Score SUS per `instrument-sus.md`. Build the TLX matrix per `instrument-nasa-tlx.md`.
20. Write Chapter 9 results. Write the deviations you decided in steps 2 and 3 into the text explicitly.

F0 - Consent and demographics

Single form, two sections. Turn **off** "collect email addresses" - it defeats the anonymity claim in the consent text.

Section 1: Consent

Paste the consent text from `consent-and-briefing.md` into the section description. Then:

Q	Type	Required	Content
1	Short answer	yes	Participant code (on your card)
2	Checkboxes	yes	`I confirm I have read and understood the above and I agree to take part`
3	Checkboxes	no	`I consent to anonymised quotations being used in the dissertation`
4	Checkboxes	no	`I agree to be contacted about a follow-up interview`
5	Checkboxes	no	`I consent to the session being screen and audio recorded` - **moderated arm only**, delete this question for the unmoderated form

Section 2: About you - G1 to G6 exactly as listed in `consent-and-briefing.md`. All multiple choice, all required except G2's "other".

F1 - Raw NASA-TLX

The one that must be right. It is submitted six times per participant.

Title it something the participant will recognise mid-task, for example "Apex - workload after each task".

Q	Type	Required	Content
1	Short answer	yes	Participant code
2	Multiple choice	yes	Which task? `Task 3 - Requirement to locked scenarios` / `Task 4 - Produce and lock the design` / `Task 6 - Break the work into implementation tasks` / `Task 7 - Test plan and QA sign-off` / `Task 8 - Deploy a different feature` / `Task 9 - Export what the tool produced`
3	Linear scale 0-10	yes	**Mental Demand**
4	Linear scale 0-10	yes	**Physical Demand**

Q	Type	Required	Content
5	Linear scale 0-10	yes	**Temporal Demand**
6	Linear scale 0-10	yes	**Effort**
7	Linear scale 0-10	yes	**Frustration**
8	Linear scale 0-10	yes	**Performance** - reversed anchors, placed last on purpose
9	Paragraph	no	Anything you want to say about this task

Every scale question needs its full description in the question help text, copied from `instrument-nasa-tlx.md`. Nobody is present to explain them.

Labels on each scale:
- Q3 to Q7: left `Very low`, right `Very high`.
- Q8 Performance: left **`Perfect`**, right **`Failure`**. Put the word "reversed" in the help text and repeat that good performance is the left end.

Task 6 and Task 8 need an extra prompt. Add a paragraph question shown conditionally, or simply add to Q9's help text:

> If you are answering for Task 6, name the task you think must be done first and say why in one sentence. If you are answering for Task 8, describe in your own words what happened when you tried to deploy.

Simplest robust option: two extra optional paragraph questions, "For task 6 only" and "For task 8 only". Conditional section logic in Google Forms is fragile and easy to misconfigure.

Scoring reminder: multiply every 0-10 response by 10 at analysis time, and write the deviation into Chapter 9.

F2 - SUS

Ten items, in order, exactly as in `instrument-sus.md`. Do not reword, reorder, add or drop any item.

Q	Type	Required	Content
1	Short answer	yes	Participant code
2-11	Linear scale **1-5**	yes	SUS items 1 to 10, verbatim

Scale labels: left `Strongly disagree`, right `Strongly agree`. Same labels on all ten - alternating them is a common and fatal build error, because the scoring formula already handles the polarity reversal.

Form description: the participant instructions block from `instrument-sus.md`.

Two separate forms, one per language. F2-EN uses the English items. F2-PT uses the validated European Portuguese items from Martins et al. (2015), which are now transcribed verbatim in `instrument-sus.md` under "As administered". Copy that table, not the "Published items, verbatim" one above it - the administered version is the same wording with `este produto` replaced by `o Apex`, which is the only permitted change.

Do not mix languages within one form. PT anchors: `Discordo totalmente` and `Concordo totalmente`.

F3 - Apex UX

Q	Type	Required	Content
1	Short answer	yes	Participant code
2-19	Multiple choice, 6 options	yes	The 18 items A1-A4, B1-B5, C1-C4, D1-D2, E1-E3
20	Paragraph	yes	What was the single most confusing thing about using Apex?
21	Paragraph	yes	Was there a point where you did not trust what the tool had produced? What made you doubt it?
22	Paragraph	yes	If you could change one thing about the interface, what would it be?

Use multiple choice, not linear scale, for Q2-Q19. The six options are

`1 Strongly disagree`, `2`, `3`, `4`, `5 Strongly agree`, `N/A - did not use this`. A linear scale cannot carry an N/A option, and forcing a centre point for a surface the participant never opened manufactures data.

Group them into four sections matching A, B, C, D and E so the form does not read as a wall of 18 statements.

Response-sheet hygiene

- Participant code is the join key across all four sheets. If a code is missing or mistyped, that row is orphaned. Print the code large on the card and ask for it first on every form.
- F1 produces six rows per participant. Sort by code then task before building the matrix.
- Google Forms writes the question text as the column header, which is unwieldy. Add a second header row of short names (`mental`, `physical`, `temporal`, `effort`, `frustration`, `performance`) before analysis, or rename the columns in a copy of the sheet. Never edit the raw response sheet itself.
- Export to CSV and keep the raw exports untouched. Do all cleaning in a copy.

What goes in Appendix B

Once the study has run, Appendix B needs: the SUS form as administered, the Raw TLX form as administered including subscale descriptions and the response scale actually used, the UX questionnaire, both interview guides, the consent form, and the per-participant raw scores in anonymised form.

Everything except the raw scores already exists in these files. Appendix B is currently four `\todo` prompts.

Apex evaluation plan - SUS, NASA-TLX and UI/UX instruments

Working document. Defines the empirical study that feeds Chapter 9 (Evaluation) and Appendix B. Written **before** running the study on purpose: pre-registering the task set, the measures and the analysis is what separates a demonstration from an anecdote, and it is the single cheapest defence against "the observations were selected after the fact to suit the artefact".

Status: **draft, not yet executed.** Nothing here has been administered.

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1. What each instrument is allowed to claim

This is the sharpest attack surface in the whole thesis, so it is stated first and never softened later.

Instrument	Artefact evaluated	Claim it supports	Claim it does NOT support
SUS	Apex (instantiation)	Perceived usability of the tool	That the framework is sound
NASA-TLX	Apex (instantiation)	Perceived workload while operating the tool	Delivered software quality
Apex UX questionnaire (custom)	Apex (instantiation)	Where specifically the UI helps or obstructs; AI trust calibration	Anything generalisable - unvalidated instrument, descriptive only
Task-performance measures	Apex (instantiation)	Objective completion, time, assists, errors	Long-term productivity
Practitioner interviews	Framework	Practicality and perceived value in real workflows	Usability of the tool
Agile/Scrum expert interviews	Framework	Alignment with and departure from established methodology	Adoption outcomes
Analytical assessment	Framework	Degree to which stated design criteria are met	Independent validation

A good SUS score for Apex is **not** evidence that the framework works. Chapter 9 already says this in `\Cref{tab:eval_instruments}`; keep it in the running text too.

Objective measures are included deliberately. Chapter 9's threats section already concedes that "the instruments measure perception rather than delivered software quality". Adding completion rate, time on task and assist count costs nothing during the session and blunts that threat instead of merely admitting it.

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2. Study design

- **Type:** ex post, naturalistic, single-setting, within-subject (no control group). Pries-Heje et al. classification already cited in Chapter 9.
- **Unit of analysis:** one practitioner completing a fixed task set in Apex on a real project.
- **Setting:** the partner organisation already using Apex (the same one driving the Plane.so migration), plus, if the sample is short, MEIC students or practitioners with a development background.
- **Session length:** about 60 minutes unmoderated, about 80 minutes moderated.
- **No control condition.** There is no baseline "same task without Apex" arm, because the task set is defined in terms of Apex's own artefacts. This is a real limitation and belongs in `\Cref{sec:threats}`, not hidden.

Two arms: unmoderated primary, moderated secondary

The study runs in two arms. This is a deliberate split, not indecision, and it must be described as such in Chapter 9.

	Arm 1 - unmoderated	Arm 2 - moderated
Target N	10-12	3-4
Format	Self-administered task script, remote, participant's own time	In person or screen-shared, facilitator present
Materials	`task-script.md` plus three online forms	Same script, plus `observer-sheet.md`
Yields	SUS, per-task Raw TLX, UX questionnaire, self-reported outcomes	All of the above **plus** think-aloud, assist counts, time on task, the Task 8 comprehension score
Does not yield	anything observational	N, realistically
Facilitator bias	none	present, mitigated only

Why both. Arm 1 is the only realistic route to a usable SUS N, and it removes the facilitator-bias problem entirely, which matters unusually much here because the author designed both artefacts. Arm 2 supplies the observational findings that a questionnaire cannot reach: where people actually got stuck, what they said while stuck, and whether they understood the Task 8 refusal. Neither arm alone produces a defensible chapter.

Report the arms separately wherever the instrument differs between them (see the TLX response-scale note below). Pooling arms silently is the error to avoid.

Per-task NASA-TLX

Raw TLX is administered **after each marked task**, not once at the end. Six administrations per participant, on script tasks 3, 4, 6, 7, 8 and 9. Rationale, costs and the anchoring/fatigue trade-off are in `instrument-nasa-tlx.md`.

The primary result becomes a six-subscale by six-task matrix, which is what lets the chapter say *where in the framework* the workload sits rather than only how much of it there is.

Session timetable

Arm 1, self-paced, roughly 60 minutes:

Approx. min	Activity
0-5	Consent form and demographics, online
5-10	Read "Before You Begin", open the three form links
10-55	**Tasks 1-9**, with the TLX form after tasks 3, 4, 6, 7, 8, 9
55-58	SUS form
58-63	Apex UX form and the three open questions

Arm 2 adds a 5-minute scripted orientation before the tasks and runs about 80 minutes, because think-aloud slows everything down.

Interviews are a **separate** session for both arms, 30-45 minutes. Running one straight after an hour of task work produces fatigue answers.

Order matters at the end: SUS **before** the UX questionnaire. SUS is the standardised, comparable instrument and should not be answered after 18 bespoke items have primed the participant about specific features.

Participants

- **Target N = 12** for SUS. Twelve is the point where SUS is generally treated as yielding a stable mean; below that, report the distribution and individual scores and make no inferential claim.
- **Realistic N is likely lower.** If it is, say so plainly, report every individual score, plot the distribution, and drop all inferential statistics. A defensible small-N descriptive report beats an indefensible t-test.
- Interviews: 5-8 practitioners plus 2-3 Agile/Scrum experts. Interviews saturate much earlier than questionnaires; the two do not need the same N.
- Recruitment, consent, anonymisation and data handling: see `consent-and-briefing.md`.

Researcher role

The author designed both artefacts. This is a conflict and must be named in `\Cref{sec:demo_design}` and again in `\Cref{sec:threats}`.

Arm 1 removes it structurally. No facilitator is present, the script is fixed text, and the forms are submitted without the author watching. This is the strongest single argument for making the unmoderated arm the primary one, and it should be stated in the chapter as a design choice rather than a convenience.

Arm 2 reintroduces the conflict, and there only mitigations are available:

1. The facilitator reads from a fixed script (in `consent-and-briefing.md`) and does not demonstrate, hint or defend the tool during tasks.
2. Assists are logged rather than silently given (observer sheet).
3. Questionnaires are self-administered, on the same forms as arm 1, without the facilitator watching.
4. Participants are told explicitly that the tool is being tested, not them, and that negative answers are the useful ones.

None of this removes the bias in arm 2. Say so, and report which findings come from which arm.

3. Task set

Participant-facing wording: `task-script.md`. What each task is testing, success criteria, expected times and the environment reset: `task-set.md`.

Nine tasks, identical for every participant, roughly 45 minutes of tool time.
Six of them carry a Raw TLX form.

#	Task	TLX	Apex surface exercised
1	Sign in and open the project		Auth, PM tool selector, project picker
2	Connect the project to GitHub		Sidebar, PAT entry, connected state
3	Requirement to locked Gherkin	yes	Phase 1, clarifying Q&A, assumptions, lock
4	Produce and lock the design	yes	Phase 2, technical + visual + runtime spec
5	Resume a previous session		Session store, active-project persistence
6	Break the work into implementation tasks	yes	Phase 3, packs, task DAG
7	Test plan and QA sign-off	yes	Phase 4
8	Deploy a **different**, un-QA'd feature	yes	Phase 5 deployment gate
9	Export what the tool produced	yes	Export and download paths

Optional and excluded from all measures: Phase 6 drift check, traceability explorer, Autopilot.

Task 8 is designed to be refused. The gate blocks a story that is not `qa_passed`. Whether the participant understands *why* is the single most informative observation available, and it is the one place the UI has to explain a refusal rather than celebrate a success. In arm 1 it is captured by the open text on the TLX form and by UX item A4; in arm 2 it is additionally scored 2/1/0 on the observer sheet.

Task 5 is a real regression test, not filler. Session state lives in `sessionStorage` and the active-project-survives-a-fresh-session bug was fixed only recently. Putting a tab close in the middle of the script exercises exactly that path with a real user, which no automated test does.

4. Instruments

File	Contents
`task-script.md`	**Participant-facing document**, EN + PT. Hand this out. Format follows the LiteraFlow task script: numbered tasks, `[TLX]` markers, reference screenshots, explicit stop instructions
`instrument-sus.md`	SUS, 10 items, EN + PT, scoring procedure, interpretation bands
`instrument-nasa-tlx.md`	Raw NASA-TLX, 6 subscales, EN + PT, subscale definitions as shown to participants
`instrument-apex-ux.md`	Custom 18-item Apex UX questionnaire + 3 open questions, EN + PT
`interview-guides.md`	Practitioner guide and Agile/Scrum expert guide
`observer-sheet.md`	Per-task completion, time, assists, errors, critical incidents
`consent-and-briefing.md`	Informed consent, facilitator script, demographics form

Raw TLX, not weighted TLX

Decision: use Raw TLX (unweighted). Justification for the chapter:

- The weighted procedure adds 15 pairwise comparisons. Under per-task administration that is 15 comparisons **times six tasks**, which is not survivable in a 60-minute unmoderated session.
- Raw TLX correlates highly with weighted TLX and is the dominant form in software-tool evaluation.
- The weights would have to be either collected once and applied to all six tasks, which assumes the participant's priorities do not change between phases, or collected six times, which nobody will complete. Neither is attractive; Raw TLX avoids the choice.

Chapter 9's `\todo` at `sec:tlx` says the justification matters more than the choice. This is the justification. Report it as "Raw TLX" explicitly, never as "NASA-TLX" unqualified.

The custom questionnaire is not a validated instrument

SUS cannot see the things this thesis actually cares about: whether the practitioner trusted the AI output, whether the phase gates read as guardrails or as obstruction, whether reviewing generated work felt cheaper than writing it. Those need bespoke items.

Consequence: report the 18 items **descriptively, item by item**, with no composite score, no Cronbach's alpha, no factor claims. An unvalidated instrument reported as a single number is exactly the sort of thing that gets attacked in a viva. Reported as a per-item diverging bar chart it is honest and genuinely informative.

Language

Every instrument is written EN + PT because the app itself ships EN/PT parity and the likely participants are Portuguese speakers.

PT SUS: decided 2026-08-11 - use the validated European Portuguese version (Martins et al., 2015). Consequences:

- **Done 2026-08-11: the paper was obtained and the ten items are transcribed verbatim in `instrument-sus.md`.** F2-PT can be built.
- Chapter 9 states that the validated European Portuguese SUS was used, and cites Martins et al. alongside Brooke.
- Only the system name is substituted ("o Apex" for "o sistema"). Nothing else changes.
- The English SUS stays available for participants who choose EN. Report which language each participant used; do not pool EN and PT into one mean without saying so, since they are strictly two instruments.
- **State the instrument's reported limits in Chapter 9 pre-emptively.** Martins et al. report construct validity (r = 0.70 vs PSSUQ) but **weak inter-rater reliability, ICC = 0.36**, and 76.67% agreement against their own 80% threshold. They attribute it to the alternating item polarity. Their sample was also drawn from the general community, not software practitioners. All of this is in `instrument-sus.md`; one sentence in the text closes the line of attack that citing a validated instrument without knowing its limits would otherwise open.

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5. Analysis

SUS

- Score per participant: odd items `response - 1`, even items `5 - response`, sum, multiply by 2.5. Range 0-100.

- **Not a percentage.** State this in the chapter; it is the most common misreading of SUS.
- Report: every individual score, mean, median, SD, min/max, and a dot plot.
- Interpret against the Bangor adjective ratings and the Sauro/Lewis curved grade, never against a bare "68 is average".

Raw TLX
- Primary artefact: the **six subscales by six tasks matrix**, as a heatmap. This is the headline figure of the chapter.
- Then per-task aggregate, so the phases can be ranked by total workload.
- Then the subscale profile collapsed across tasks, for the overall claim.
- The single grand aggregate last, if at all.
- Per-task predictions are written down in `instrument-nasa-tlx.md` in advance, including the specific prediction that task 8, the deployment gate, carries the highest Frustration and lowest Performance. A mismatch is then a finding rather than something rationalised afterwards.
- If the two arms used different response scales, report them separately and do not pool them into one mean without saying so.

Apex UX questionnaire
- Per-item stacked/diverging bars. No composite.

Task performance
- Arm 1: self-reported completion, plus form timestamps as a rough proxy for time per task. Treat the timestamps as indicative only; an unmoderated participant can walk away mid-task.
- Arm 2: completion rate, median time on task, assist count, error count, and the task 8 comprehension score from the observer sheet.
- Any task under 100% completion gets a written explanation, not just a number.

Interviews
- Thematic analysis, codebook developed from the first two transcripts then applied to the rest.
- Report disconfirming evidence explicitly and with the same prominence.

Triangulation
- Cross-read the four sources. Where they disagree, the disagreement *is* the finding. Example worth watching for: a good SUS score alongside interview scepticism about process overhead - that pattern says the tool is pleasant and the method is heavy, which is a real and publishable result.

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6. What still has to be decided

1. **RQ naming.** `\Cref{tab:eval_instruments}` maps instruments to RQ-A..RQ-D, which are still the unresolved assistant-written set colliding with the SLR's RQ1-RQ5. Every instrument mapping in this plan inherits that. Resolve the naming before Appendix B is typeset.
2. ~~Validated PT SUS or own translation.~~ **CLOSED 2026-08-11: validated European Portuguese version (Martins et al., 2015), obtained and transcribed.** Nothing outstanding.
3. **N and recruitment.** Whether the partner organisation can supply 12, or whether the sample gets topped up with MEIC students, which changes the external-validity claim.
4. **Ethics approval.** Whether IST requires a formal submission for this study or whether informed consent plus anonymisation suffices. Ask the supervisor.
5. **Arm split.** The 10-12 unmoderated plus 3-4 moderated split above is a proposal. Confirm the moderated arm is worth the scheduling cost, or drop it and accept losing the observational findings.
6. **TLX response scale.** Google Forms cannot render the 20-interval line. The recommendation is a 0-10 scale rescaled to 0-100, documented as a deviation. Alternative: use a form tool with a real slider. Decide before the pilot, and never change it mid-study.
7. **Form platform.** Google Forms is assumed. Anything with a 0-100 slider and CSV export is better. Whatever is chosen must let a participant submit the TLX form six times without re-entering their code by hand each time, or use a prefilled link per task.
8. **Reference screenshots.** `task-script.md` has nine `[SCREENSHOT: ...]` slots that must be captured from a real seeded project. In an unmoderated study these are load-bearing, not decoration: without them a participant can spend ten minutes doing the wrong thing with nobody there to notice.

7. Missing bibliography entries

None of these exist in `Bibliography.bib` yet. All are required before Chapter 9 can cite the instruments.

```
@incollection{Brooke:1996SUS,
    Author = {Brooke, J.},
    Booktitle = {Usability Evaluation in Industry},
    Editor = {Jordan, P. W. and Thomas, B. and Weerdmeester, B. A. and McClelland, I. L.},
    Pages = {189--194},
```

```
    Publisher = {Taylor and Francis},
    Title = {{SUS: A `quick and dirty' usability scale}},
    Year = {1996}
}

@article{Bangor:2009SUS,
    Author = {Bangor, A. and Kortum, P. and Miller, J.},
    Journal = {Journal of Usability Studies},
    Number = {3},
    Pages = {114--123},
    Title = {{Determining what individual SUS scores mean: Adding an adjective
rating scale}},
    Volume = {4},
    Year = {2009}
}

@book{Sauro:2016QUANT,
    Author = {Sauro, J. and Lewis, J. R.},
    Edition = {2nd},
    Publisher = {Morgan Kaufmann},
    Title = {{Quantifying the User Experience: Practical Statistics for User
Research}},
    Year = {2016}
}

@incollection{Hart:1988TLX,
    Author = {Hart, S. G. and Staveland, L. E.},
    Booktitle = {Human Mental Workload},
    Editor = {Hancock, P. A. and Meshkati, N.},
    Pages = {139--183},
    Publisher = {North-Holland},
    Title = {{Development of NASA-TLX (Task Load Index): Results of empirical and
theoretical research}},
    Year = {1988}
}

@inproceedings{Hart:2006TLX,
    Author = {Hart, S. G.},
    Booktitle = {Proceedings of the Human Factors and Ergonomics Society Annual
Meeting},
    Pages = {904--908},
    Title = {{NASA-Task Load Index (NASA-TLX); 20 years later}},
    Volume = {50},
    Year = {2006}
}
```

```
@article{Lewis:2009SUS,
    Author = {Lewis, J. R. and Sauro, J.},
    Journal = {Human Centered Design, HCII 2009, LNCS},
    Pages = {94--103},
    Title = {{The factor structure of the System Usability Scale}},
    Volume = {5619},
    Year = {2009}
}
```

Required by the PT SUS decision:

```
@article{Martins:2015SUS,
    Author = {Martins, A. I. and Rosa, A. F. and Queir\'os, A. and Silva, A. and
Rocha, N. P.},
    Journal = {Procedia Computer Science},
    Pages = {293--300},
    Title = {{European Portuguese Validation of the System Usability Scale
(SUS)}},
    Doi = {10.1016/j.procs.2015.09.273},
    Url = {https://doi.org/10.1016/j.procs.2015.09.273},
    Volume = {67},
    Year = {2015}
}
```

Authors, volume, pages, year and DOI all verified against the published PDF.
Note the article title capitalises "validation" lowercase in the original.

8. Pre-flight checklist

- [ ] RQ naming resolved, `tab:eval_instruments` updated
- [x] PT SUS decided: validated European Portuguese version, Martins et al. 2015
- [x] Martins et al. obtained; the ten PT items transcribed verbatim into `instrument-sus.md` and the provisional translation deleted
- [ ] Bibliography entries above added and the build reverified
- [ ] Supervisor sign-off on the task set and the session length
- [ ] Ethics/consent route confirmed with the supervisor
- [ ] Arm split confirmed, or the moderated arm dropped
- [ ] TLX response scale decided and the deviation wording drafted
- [ ] Three forms built: TLX (with the task selector and participant code first), SUS, UX. Test-submit each one and check the CSV export column names
- [ ] All nine reference screenshots captured from a seeded project
- [ ] `task-script.md` exported to the participant-facing format with the four

placeholder links filled in
- [ ] Participant code cards printed
- [ ] Pilot run with **one** participant, not counted in results, to time the script and catch broken instructions. Six TLX submissions must come back correctly labelled - if they do not, fix the form before anyone else runs
- [ ] A clean demo project seeded, including the un-QA'd `Export board to CSV` story that task 8 depends on, and a reset procedure that actually works
- [ ] Chapter 8 `sec:demo_design` written before the first real session

Task set - internal notes

Not handed to participants. The participant-facing wording is `task-script.md`; this file records what each task is actually testing, how to judge it, and how to prepare the environment.

Nine tasks, roughly 45 minutes of tool time. Six carry a Raw TLX form.

Environment, reset before every participant

- A dedicated **Demo Project** on the PM instance, seeded with 3 epics and 6 stories at `phase_status: new`, plus one story named **Export board to CSV** that must stay un-QA'd. Task 8 depends entirely on that story existing and never having passed QA.
- A GitHub repo with a **real codebase** in it, not an empty repo. Phase 3 and Phase 4 grounding read `github-context.md`; an empty repo produces vacuous output and the participant is then evaluating nothing.
- A PAT and repo URL printed on the participant card, along with sign-in credentials and the participant code.
- AI provider key configured server-side so nobody is blocked on it.
- Language left at the participant's choice from the start, then not changed.

Reset procedure: delete `contextspec/<instance_id>/<project_id>/` for the demo project and re-seed the PM board. Verify the reset by loading the project and confirming every story is back at `new` and `Export board to CSV` has no QA record.

Test the reset before the pilot. A half-reset project is the most likely way to silently ruin a participant's data, because task 8 will not refuse if the story already passed QA in a previous session.

Per-task notes

Task 1 - Sign in and open the project

Surface: auth, PM tool selector, project picker.
Success: signed in, correct project active.
Watch for: whether the PM tool selector on the login form is understood at all. It was only recently labelled.
Expected: 3 min. No TLX.

Task 2 - Connect the project to GitHub

Surface: sidebar GitHub section, PAT entry, connected state.
Success: repo shows as connected.
Watch for: whether the connected state is legible after connecting; whether the participant looks for GitHub in Settings rather than the sidebar.
Expected: 4 min. No TLX.

Task 3 `[TLX]` - Requirement to locked Gherkin

Surface: Phase 1, clarifying Q&A, per-scenario assumptions, lock.
Success: functional spec written, questions answered, Gherkin compiled, story at `gherkin_locked`.
Watch for: whether the clarifying questions are read or clicked through; whether the amber assumptions box is noticed; whether the participant hesitates before locking. Also whether a 30-60 s generation makes them think it hung.
Predicted TLX: highest Mental Demand of the session.
Expected: 10 min.

Task 4 `[TLX]` - Produce and lock the design

Surface: Phase 2, technical spec, runtime spec, visual design, lock.
Success: all three generated, assumptions reviewed, story at `design_locked`.
Watch for: whether the Visual Design group is found at all - it is collapsed by default; whether the runtime spec is read or scrolled past; **whether anything is edited or the output is accepted wholesale.** Blanket acceptance is

a trust-calibration finding and is the one this thesis most needs.
Predicted TLX: high Mental Demand and high Effort.
Expected: 10 min.

Task 5 - Resume a previous session

Surface: `sessionStorage` session store, active-project persistence, GitHub auto-restore from the encrypted server-side PAT.
Success: signs back in and lands on the same project and work.
Why it is here: this is a real regression path. Session state is in `sessionStorage`, and the "active project does not survive a fresh session" bug was fixed only recently. No automated test exercises it with a human.
Watch for: whether GitHub is still connected afterwards, or whether the participant has to reconnect. If they do, that is a bug found by the study.
Expected: 3 min. No TLX.

Task 6 `[TLX]` - Break the work into implementation tasks

Surface: Phase 3 packs, task DAG, effort and coverage metadata.
Success: packs generated, and the participant names a defensible root task with a reason.
Grading the written answer: correct if it names a task with no unmet dependencies and the justification refers to ordering or dependency rather than to the task sounding important.
Watch for: whether the DAG is read as a dependency order or as decoration.
Expected: 8 min.

Task 7 `[TLX]` - Test plan and QA sign-off

Surface: Phase 4.
Success: test plan generated, QA recorded, story at `qa_passed`.
Watch for: whether the participant grasps that QA sign-off is a human decision the tool deliberately will not make for them.
Predicted TLX: lowest demands of the marked tasks.
Expected: 6 min.

Task 8 `[TLX]` - The deployment gate

Surface: Phase 5 gate, phase-status progression, refusal messaging.
Designed to be refused. `Export board to CSV` is not `qa_passed`, so the gate blocks it.

The measure is comprehension, not completion. Score the written explanation:

- **2** - names the missing QA step and the required order
- **1** - knows a process rule blocked them, cannot say which
- **0** - concludes the tool is broken or the story is corrupt

Watch for: whether they hunt for an override; whether they find the phase status; the exact words they use for the refusal.
Predicted TLX: highest Frustration, lowest Performance. If Frustration is only moderate, the refusal messaging is doing its job, and that is a reportable result about error surfacing.
Expected: 6 min.

Task 9 `[TLX]` - Export

Surface: export and download paths, zip packaging.
Success: files downloaded and opened.
Watch for: whether the export control is findable; whether anything downloads empty or corrupt.
Predicted TLX: lowest overall.
Expected: 5 min.

Optional, excluded from all measures

- **O1** Phase 6 drift check. **O2** traceability explorer. **O3** Autopilot.

O2 is worth pushing for where time allows: traceability is a distinctive claim of the framework and has never been usability-tested.

Facilitator rules, arm 2 only

Arm 1 has no facilitator. In the moderated arm:

- Read each task aloud from `task-script.md`, verbatim. Do not paraphrase.
- Do not point, hover, or say "it's over there". Silence is data.
- Stuck for **90 seconds**: offer one hint, log it as an assist. Another 90 seconds: second hint. After a third, mark the task failed and move on.
- Prompt "what are you thinking?" only after 20 seconds of silence. Never "what would you expect to happen?" - that leads.
- Never explain, defend or apologise for the tool. Note the complaint, move on.

- The participant still fills in the same online forms, unwatched.

Task Script - Apex

Participant-facing document. Format follows the LiteraFlow task script: a self-administered, unmoderated sequence with a short NASA-TLX form after each substantive task and a single SUS at the end.

Source for the Word or Google Docs version handed to participants. Keep the formatting: numbered tasks, a `[TLX]` marker on tasks that require a form, a reference screenshot per task, and an explicit stop instruction after each marked task.

Marker convention: `[TLX]` in place of a star symbol. Prints reliably in every export path and screen reader, and cannot be mistaken for a rating.

Before handing this out:
- Replace `<TOOL LINK>`, `<TLX FORM LINK>`, `<SUS FORM LINK>`, `<UX FORM LINK>`.
- Capture the reference screenshots (slots marked below). They are not optional: in an unmoderated study they are the only thing preventing a participant from silently doing the wrong thing for ten minutes.
- Run one pilot participant who is not counted.

EN version

Task Script

Tool link: `<TOOL LINK>`

Before You Begin

Thank you for taking part in this usability study. I am evaluating Apex, a tool that supports a process for building software with AI assistance. **I am testing**

the tool, not you. There are no right or wrong answers. If something is confusing or does not work as you expected, that is exactly the feedback I need.

The session takes about 60 minutes in total.

Please read carefully before starting

- Complete all tasks in the order listed. Do not skip tasks.
- Tasks marked with `[TLX]` require you to fill in a short form immediately after finishing that task, **before** reading the next one.
- The tool uses AI. Some steps take up to a minute to finish. This is normal.
- If you get completely stuck on a task for more than five minutes, move on to the next one and note it in the final form.
- After the very last task, fill in the SUS form and then the short UX form.

Your form links - keep this document open throughout

Form	Link
NASA-TLX form (after every `[TLX]` task)	`<TLX FORM LINK>`
SUS form (end of session)	`<SUS FORM LINK>`
Apex UX form (end of session)	`<UX FORM LINK>`

Your participant code is on the card you were given. Enter it at the top of every form so the answers can be linked together. Your name is never recorded.

Tasks

Task 1 - Sign in and open the project

Sign in to Apex with the credentials on your card, and open the project called **Demo Project**.

Reference screenshot: `[SCREENSHOT: sign-in form with the PM tool selector]`

`->` Continue to the next task without filling in any form.

Task 2 - Connect the project to GitHub

Connect the project to the GitHub repository listed on your card, using the access token on the same card.

Reference screenshot: `[SCREENSHOT: sidebar GitHub section, connected state]`

`->` Continue to the next task without filling in any form.

Task 3 `[TLX]` - Turn a requirement into locked scenarios

The team needs a new feature: **users must be able to reset a forgotten password by email.**

Using Phase 1, write that requirement in your own words, answer any questions the tool asks you, produce the Gherkin scenarios, and lock them.

Reference screenshot: `[SCREENSHOT: Phase 1 with generated Gherkin and the assumptions panel visible]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 3 before continuing.**

Task 4 `[TLX]` - Produce and lock the design

Continue with the same feature into Phase 2. Generate the technical design and the visual design, review what the tool assumed on your behalf, change anything you disagree with, and lock the design.

Reference screenshot: `[SCREENSHOT: Phase 2 showing Technical Design and the collapsed Visual Design group]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 4 before continuing.**

Task 5 - Resume a previous session

Close the browser tab completely, open the tool link again, sign back in, and return to the work you were doing instead of starting something new.

Reference screenshot: `[SCREENSHOT: returning to the same active project after a fresh sign-in]`

`->` Continue to the next task without filling in any form.

Task 6 `[TLX]` - Break the work into implementation tasks

Generate the implementation packs for the password reset feature. Then look at the task breakdown and decide which task would have to be done first.

Write the name of that task, and one sentence saying why, in the NASA-TLX form when you open it.

Reference screenshot: `[SCREENSHOT: Phase 3 packs with the task dependency graph]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 6 before continuing.**

Task 7 `[TLX]` - Test plan and QA sign-off

Generate a test plan for the password reset feature, review it, and record that QA has passed.

Reference screenshot: `[SCREENSHOT: Phase 4 test plan with the QA sign-off control]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 7 before continuing.**

Task 8 `[TLX]` - Deploy a different feature

Now try to deploy a **different** story, the one called **Export board to CSV**.

Tell the form what happened, in your own words.

Reference screenshot: `[SCREENSHOT: Phase 5 deployment gate]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 8 before continuing.**

Task 9 `[TLX]` - Export what the tool produced

Export the artefacts the tool has produced for the password reset feature. Download them and confirm that the files opened correctly.

Reference screenshot: `[SCREENSHOT: export or download-all control]`

`->` **Stop here. Open the NASA-TLX form and fill it in for Task 9 before continuing.**

Optional - only if you have time and interest

Not required, and not part of the study measures. Nothing to fill in.

- **O1** - Open Phase 6 and find out whether any feature has drifted from what was specified.
- **O2** - Find every artefact that traces back to the password reset requirement.
- **O3** - Run Autopilot on one epic and stop it before it finishes.

You have completed all tasks

Please now open the **SUS form** (`<SUS FORM LINK>`) and fill it in. This takes about three minutes.

Then open the **Apex UX form** (`<UX FORM LINK>`), which takes about five minutes and includes three open questions.

Thank you for your time and feedback.

Versão PT

Guião de Tarefas

Ligação para a ferramenta: `<TOOL LINK>`

Antes de Começar

Obrigado por participar neste estudo de usabilidade. Estou a avaliar o Apex, uma ferramenta que apoia um processo de desenvolvimento de software com apoio de IA. **Estou a testar a ferramenta, não o participante.** Não há respostas certas ou erradas. Se algo for confuso ou não funcionar como esperava, é exactamente esse o contributo de que preciso.

A sessão demora cerca de 60 minutos no total.

Leia com atenção antes de começar

- Realize todas as tarefas pela ordem indicada. Não salte tarefas.
- As tarefas marcadas com `[TLX]` exigem o preenchimento de um formulário curto imediatamente a seguir, **antes** de ler a tarefa seguinte.
- A ferramenta usa IA. Alguns passos demoram até um minuto. É normal.
- Se ficar completamente bloqueado numa tarefa durante mais de cinco minutos, passe à seguinte e indique-o no formulário final.
- Depois da última tarefa, preencha o formulário SUS e depois o formulário UX.

As suas ligações - mantenha este documento aberto

Formulário	Ligação
Formulário NASA-TLX (após cada tarefa `[TLX]`)	`<TLX FORM LINK>`
Formulário SUS (fim da sessão)	`<SUS FORM LINK>`
Formulário UX do Apex (fim da sessão)	`<UX FORM LINK>`

Formulário	Ligação

O seu código de participante está no cartão que recebeu. Introduza-o no início de cada formulário para que as respostas possam ser ligadas entre si. O seu nome nunca é registado.

Tarefas

Tarefa 1 - Entrar e abrir o projecto

Entre no Apex com as credenciais do seu cartão e abra o projecto chamado **Demo Project**.

`->` Passe à tarefa seguinte sem preencher qualquer formulário.

Tarefa 2 - Ligar o projecto ao GitHub

Ligue o projecto ao repositório GitHub indicado no seu cartão, usando o token de acesso do mesmo cartão.

`->` Passe à tarefa seguinte sem preencher qualquer formulário.

Tarefa 3 `[TLX]` - Transformar um requisito em cenários bloqueados

A equipa precisa de uma nova funcionalidade: **os utilizadores devem poder recuperar uma palavra-passe esquecida por email.**

Usando a Fase 1, escreva esse requisito por palavras suas, responda às perguntas que a ferramenta lhe fizer, produza os cenários Gherkin e bloqueie-os.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 3 antes de continuar.**

Tarefa 4 `[TLX]` - Produzir e bloquear o design

Continue com a mesma funcionalidade para a Fase 2. Gere o design técnico e o design visual, reveja o que a ferramenta assumiu por si, altere aquilo de que discordar e bloqueie o design.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 4 antes de continuar.**

Tarefa 5 - Retomar uma sessão anterior

Feche completamente o separador do browser, abra novamente a ligação da ferramenta, volte a entrar e retome o trabalho onde estava, em vez de começar algo novo.

`->` Passe à tarefa seguinte sem preencher qualquer formulário.

Tarefa 6 `[TLX]` - Dividir o trabalho em tarefas de implementação

Gere os pacotes de implementação para a funcionalidade de recuperação de palavra-passe. Depois observe a divisão em tarefas e decida qual teria de ser feita em primeiro lugar.

Escreva o nome dessa tarefa, e uma frase a explicar porquê, no formulário NASA-TLX quando o abrir.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 6 antes de continuar.**

Tarefa 7 `[TLX]` - Plano de testes e aprovação de QA

Gere um plano de testes para a funcionalidade, reveja-o e registe que o QA foi aprovado.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 7 antes de continuar.**

Tarefa 8 `[TLX]` - Colocar em produção uma funcionalidade diferente

Agora tente colocar em produção uma funcionalidade **diferente**, a que se chama **Export board to CSV**.

Diga no formulário o que aconteceu, por palavras suas.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 8 antes de continuar.**

Tarefa 9 `[TLX]` - Exportar o que a ferramenta produziu

Exporte os artefactos que a ferramenta produziu para a funcionalidade de recuperação de palavra-passe. Descarregue-os e confirme que os ficheiros abriram correctamente.

`->` **Pare aqui. Abra o formulário NASA-TLX e preencha-o para a Tarefa 9 antes de continuar.**

Opcional - só se tiver tempo e interesse

Não é obrigatório e não faz parte das medidas do estudo. Nada a preencher.

- **O1** - Abra a Fase 6 e descubra se alguma funcionalidade se desviou daquilo que foi especificado.
- **O2** - Encontre todos os artefactos que remetem para o requisito de recuperação de palavra-passe.
- **O3** - Execute o Autopilot num épico e pare-o antes de terminar.

Concluiu todas as tarefas

Abra agora o **formulário SUS** (`<SUS FORM LINK>`) e preencha-o. Demora cerca de três minutos.

Depois abra o **formulário UX do Apex** (`<UX FORM LINK>`), que demora cerca de cinco minutos e inclui três perguntas abertas.

Obrigado pelo seu tempo e pelo seu contributo.

System Usability Scale - as administered for Apex

Source instrument: Brooke (1996). Free to use, requires acknowledgement of the source. Ten items, five-point scale, alternating positive and negative polarity.

Adaptation: the only change from the standard wording is substituting "Apex" for "the system". No item is reworded, reordered, added or dropped - doing any of that invalidates the comparability that is the entire point of using SUS. Item 8's wording is given in Brooke's original form ("cumbersome"); the widely used "awkward" variant is noted where relevant but not used here, so that the reported score stays comparable to the standard corpus.

Administer **after** Raw TLX, self-completed, unwatched.

Instructions shown to the participant (EN)

> Please answer each statement about **Apex**, the tool you have just used, by marking one box. There are no right answers. Record your immediate reaction rather than thinking for a long time about each one. If you feel you cannot answer an item, mark the centre point.

Scale: `1 = Strongly disagree` ... `5 = Strongly agree`

#	Item	1	2	3	4	5
1	I think that I would like to use Apex frequently.					
2	I found Apex unnecessarily complex.					
3	I thought Apex was easy to use.					
4	I think that I would need the support of a technical person to be able to use Apex.					
5	I found the various functions in Apex were well integrated.					
6	I thought there was too much inconsistency in Apex.					

#	Item	1	2	3	4	5
7	I would imagine that most people would learn to use Apex very quickly.					
8	I found Apex very cumbersome to use.					
9	I felt very confident using Apex.					
10	I needed to learn a lot of things before I could get going with Apex.					

Versão portuguesa validada (PT)

Source of record: Martins, A. I., Rosa, A. F., Queirós, A., Silva, A., and Rocha, N. P. (2015). *European Portuguese validation of the System Usability Scale (SUS).* Procedia Computer Science, 67, 293-300. DOI `10.1016/j.procs.2015.09.273`. Open access, CC BY-NC-ND.

Presented at DSAI 2015. The ten items below are their Table 1, transcribed verbatim. This is the instrument; the provisional translation that previously sat here has been discarded.

Note their generic referent is **"produto"**, not "sistema".

Published items, verbatim

#	Original item	Item correspondente em português
1	I think that I would like to use this system frequently.	Acho que gostaria de utilizar este produto com frequência.
2	I found the system unnecessarily complex.	Considerei o produto mais complexo do que necessário.
3	I thought the system was easy to use.	Achei o produto fácil de utilizar.
4	I think that I would need the support of a technical person to be able to use this system.	Acho que necessitaria de ajuda de um técnico para conseguir utilizar este produto.
5	I found the various functions in this system were well integrated.	Considerei que as várias funcionalidades deste produto estavam bem integradas.

#	Original item	Item correspondente em português
6	I thought there was too much inconsistency in this system.	Achei que este produto tinha muitas inconsistências.
7	I would imagine that most people would learn to use this system very quickly.	Suponho que a maioria das pessoas aprenderia a utilizar rapidamente este produto.
8	I found the system very cumbersome to use.	Considerei o produto muito complicado de utilizar.
9	I felt very confident using the system.	Senti-me muito confiante a utilizar este produto.
10	I needed to learn a lot of things before I could get going with this system.	Tive que aprender muito antes de conseguir lidar com este produto.

As administered - substituting the system name

The only change is the referent: `este produto` / `o produto` becomes `o Apex`. Nothing else moves. Same order, same polarity, same words. Declare the substitution in one sentence in Chapter 9.

Escala: `1 = Discordo totalmente` ... `5 = Concordo totalmente`

#	Afirmação	1	2	3	4	5
1	Acho que gostaria de utilizar o Apex com frequência.					
2	Considerei o Apex mais complexo do que necessário.					
3	Achei o Apex fácil de utilizar.					
4	Acho que necessitaria de ajuda de um técnico para conseguir utilizar o Apex.					
5	Considerei que as várias funcionalidades do Apex estavam bem integradas.					
6	Achei que o Apex tinha muitas inconsistências.					
7	Suponho que a maioria das pessoas aprenderia a utilizar rapidamente o Apex.					
8	Considerei o Apex muito complicado de utilizar.					
9	Senti-me muito confiante a utilizar o Apex.					

#	Afirmação	1	2	3	4	5
10	Tive que aprender muito antes de conseguir lidar com o Apex.					

Response anchors. The paper does not print Portuguese anchor labels, only that SUS is a five-point Likert scale of strength of agreement. `Discordo totalmente` and `Concordo totalmente` are the conventional wording and are what this study uses; say so rather than implying the anchors came from Martins et al.

What the validation actually established, and what it did not

State this in Chapter 9 **before** an examiner raises it. Citing a validated instrument without knowing its reported limits is worse than using a translation openly.

Property	Reported
Semantic and content equivalence	established, pilot with 4 participants
Construct validity vs PSSUQ	r = 0.70, significant
Construct validity vs a general usability question	r = 0.48, p < 0.05
Inter-rater reliability, ICC	**0.36, which the authors classify as weak** (CI 95%: 0.01 to 0.63)
Percentage of agreement	76.67%, below the 80% they state as acceptable

The authors' own conclusion, paraphrased: the Portuguese version can be used to distinguish usable from non-usable applications, but the low ICC means they intend to validate the all-positive variant of SUS and repeat the study with a larger sample. They attribute the weak reliability to the alternating positive/negative item polarity causing filling errors, and cite Sauro and Lewis on the same problem.

Two consequences for this thesis:

1. **Do not overclaim.** Write "the European Portuguese version validated by Martins et al. (2015), whose authors report construct validity but weak inter-rater reliability (ICC = 0.36)". That sentence costs nothing and closes the line of attack completely.
2. **Their sample was drawn from the general community**, not software

practitioners. The linguistic validation carries over; the reliability estimate was obtained in a different population from this study's. Worth one sentence in `\Cref{sec:threats}`.

Do **not** switch to the all-positive variant. It is not validated in European Portuguese, and adopting an unvalidated variant to dodge a caveat about a validated one is the wrong trade.

> **Instruções ao participante:** Responda a cada afirmação sobre o **Apex**, a ferramenta que acabou de utilizar, marcando uma opção. Não há respostas certas. Registe a sua reacção imediata em vez de pensar muito tempo em cada uma. Se sentir que não consegue responder, marque o ponto central.

Scoring

For each participant:

1. **Odd items (1, 3, 5, 7, 9):** contribution = `response - 1`
2. **Even items (2, 4, 6, 8, 10):** contribution = `5 - response`
3. Sum the ten contributions. Range 0-40.
4. Multiply by 2.5. **Final range 0-100.**

Worked example - a participant answering `4,2,4,2,4,2,4,2,4,2`: odd contributions `3,3,3,3,3` = 15; even contributions `3,3,3,3,3` = 15; sum 30; SUS = **75**.

The score is not a percentage

Say this explicitly in Chapter 9. A SUS of 75 does not mean "75% usable" and does not sit at the 75th percentile. It is the single most common misreading of the instrument and an examiner may well probe it.

Interpretation

Report against two published scales rather than a bare number.

Bangor et al. (2009) adjective ratings - approximate mean SUS per adjective:

Adjective	approx. SUS
Worst imaginable	12.5
Awful	20.3
Poor	35.7
OK	50.9
Good	71.4
Excellent	85.5
Best imaginable	90.9

Note that "OK" sits near 51, not near 68. A score in the fifties is not acceptable-but-fine; it is mediocre.

Sauro and Lewis curved grading scale - roughly:

Grade	SUS range
A+	84.1 and above
A	80.8 - 84.0
A-	78.9 - 80.7
B+	77.2 - 78.8
B	74.1 - 77.1
B-	72.6 - 74.0
C+	71.1 - 72.5
C	65.0 - 71.0
C-	62.7 - 64.9
D	51.7 - 62.6
F	below 51.7

Grade	SUS range

The mean SUS across the published corpus is approximately 68, which is why 68 sits at grade C.

Verify the exact band boundaries against the printed source before they go into the thesis. They are reproduced here from a secondary reading and are close enough to plan with, not to publish unchecked.

Sub-scales

Lewis and Sauro (2009) split SUS into Usability (items 1-3, 5-9) and Learnability (items 4 and 10). Reporting the Learnability sub-scale separately is worthwhile here, because Apex's phase-gate model is exactly the kind of thing that scores well on usability and badly on learnability, and that split would be a genuine finding rather than noise.

Reporting

With N around 12, report:

- every individual score, in a table, anonymised as P1..Pn
- mean, median, standard deviation, minimum, maximum
- a dot plot of the distribution
- the adjective band and letter grade of the mean
- the Usability and Learnability sub-scale means, separately

Do **not** run t-tests, confidence intervals or significance claims on a sample this size in a single setting. Descriptive reporting is defensible; inferential reporting on N=12 non-random participants is not.

Raw NASA-TLX - as administered for Apex

Source instrument: Hart and Staveland (1988); Raw TLX variant discussed in Hart (2006). Six subscales.

Variant used: Raw TLX (unweighted). The 15 pairwise weighting comparisons are **not** administered. Justification is in `EVALUATION-PLAN.md` section 4 and must appear in Chapter 9 - reporting an unweighted score as plain "NASA-TLX" is the error to avoid. Always write "Raw TLX".

Administration: per task, not per session

Decision: Raw TLX is administered once after each marked task, immediately, before the participant reads the next task. Six administrations per participant (script tasks 3, 4, 6, 7, 8, 9).

This is the instrument's intended use. NASA-TLX is a *task* load index; a single end-of-session administration is the compromise, not the standard. Three consequences, all favourable:

1. **Workload profile per phase.** The primary result becomes a matrix of six subscales by six tasks, not one flat score. "Phase 2 design carries the highest mental demand while Phase 5 carries the highest frustration" is a claim about *where in the framework* the burden sits, which is what this thesis is actually arguing about. A single aggregate cannot say that.
2. **No recall decay.** Each rating is taken within a minute of the work it describes.
3. **More data points.** Participants x 6 tasks, so the workload analysis is not hostage to the same small N as SUS.

Cost: roughly 90 seconds per administration, about 9 minutes across the session, plus the interruption. Accepted.

Frame of reference given to the participant: "the task you have just completed", named explicitly. The form's first question is which task the responses refer to; without it the whole matrix is unrecoverable.

Note the trade-off honestly in Chapter 9: repeated administration invites anchoring, where later ratings drift toward earlier ones, and fatigue on the last one or two tasks. Neither is avoidable here. State them in `\Cref{sec:threats}` alongside the benefits above.

Instructions shown to the participant (EN)

> Which task are you answering about? `(required, single choice)`
>
> Your participant code: `(required)`

> We are interested in the workload you experienced **during the task you have just completed**. Please rate each of the six aspects below by marking a point on the line. Read each description before answering. Note that the scale for Performance runs in the opposite direction to the others.

Each scale is a line divided into 20 intervals. Mark one interval. The mark is converted to a score from 0 to 100 in steps of 5.

Mental Demand

How much mental and perceptual activity was required? Was it easy or demanding, simple or complex, forgiving or exacting?

```
Very Low |--------------------| Very High
```

Physical Demand

How much physical activity was required? Was the task easy or demanding, slow or brisk, restful or laborious?

```
Very Low |--------------------| Very High
```

Temporal Demand

How much time pressure did you feel due to the rate or pace at which the tasks occurred? Was the pace slow and leisurely or rapid and frantic?

```
Very Low |--------------------| Very High
```

Performance

How successful do you think you were in accomplishing the goals of this task? How satisfied were you with your performance in accomplishing them?

```
Perfect |--------------------| Failure
```

Note the reversed anchors. Good performance is at the left end.

Effort

How hard did you have to work, mentally and physically, to accomplish your level of performance?

`Very Low |-------------------| Very High`

Frustration Level

How insecure, discouraged, irritated, stressed and annoyed, versus secure, gratified, content and relaxed, did you feel during this task?

`Very Low |-------------------| Very High`

Instruções apresentadas ao participante (PT)

> A que tarefa se refere esta resposta? `(obrigatório, escolha única)`
>
> O seu código de participante: `(obrigatório)`
>
> Interessa-nos a carga de trabalho que sentiu **durante a tarefa que acabou de realizar**. Classifique cada um dos seis aspectos abaixo marcando um ponto na linha. Leia cada descrição antes de responder. Note que a escala do Desempenho corre no sentido inverso das restantes.

Exigência Mental

Quanta actividade mental e perceptiva foi necessária? A tarefa foi fácil ou exigente, simples ou complexa, tolerante ou rigorosa?

`Muito Baixa |-------------------| Muito Alta`

Exigência Física

Quanta actividade física foi necessária? A tarefa foi fácil ou exigente, lenta ou rápida, repousante ou trabalhosa?

`Muito Baixa |-------------------| Muito Alta`

Exigência Temporal

Quanta pressão de tempo sentiu devido ao ritmo a que a tarefa decorreu? O ritmo foi lento e tranquilo ou rápido e frenético?

`Muito Baixa |-------------------| Muito Alta`

Desempenho

Em que medida acha que teve sucesso em atingir os objectivos desta tarefa? Quão satisfeito ficou com o seu desempenho?

```
Perfeito |--------------------| Fracasso
```

Note as âncoras invertidas. O bom desempenho está na extremidade esquerda.

Esforço

Quanto teve de trabalhar, mental e fisicamente, para alcançar o seu nível de desempenho?

```
Muito Baixo |--------------------| Muito Alto
```

Nível de Frustração

Quão inseguro, desencorajado, irritado, tenso e incomodado, por oposição a seguro, satisfeito, contente e tranquilo, se sentiu durante esta tarefa?

```
Muito Baixo |--------------------| Muito Alto
```

Building this as an online form

The script is unmoderated, so the TLX is a web form the participant opens six times. Two things must be got right or the data is unusable.

1. Which task is this for

First question on the form, required, single choice: Task 3, Task 4, Task 6, Task 7, Task 8, Task 9, with the task titles spelled out. Second question: participant code, required, short answer.

Without both, six submissions per participant cannot be reassembled into a matrix, and one participant forgetting their code loses their whole row.

2. The response scale is a real deviation from the instrument

The published NASA-TLX scale is a line divided into **20 intervals**, scored 0-100 in steps of 5.

Google Forms cannot reproduce this. Its linear scale tops out at 10 points and it has no slider. The options, in order of preference:

Option	Fidelity	Verdict
A form tool with a real 0-100 slider	faithful	best if one is available
Google Forms linear scale 0-10, multiplied by 10	coarse but monotonic	acceptable, **must be documented**
Google Forms short answer, number 0-100 validated as a multiple of 5	faithful values	high error rate, participants type freely
Google Forms grid with 21 columns	faithful	unusable on a phone, do not

Recommendation: 0-10 linear scale, multiplied by 10 at analysis time. Then write in Chapter 9, in plain words: *"a 0-10 response scale was used in place of the original 20-interval line, and scores were rescaled to 0-100; this reduces the resolution of the instrument and the resulting values are not directly comparable to studies using the original scale."*

Do not silently rescale and report it as NASA-TLX. That is the sort of quiet deviation that, once noticed in a viva, casts doubt on everything else.

3. Anchors on every item

Repeat the subscale description and both anchor words on every question. In an unmoderated study nobody is present to explain that Performance runs backwards, and a participant who misses that inverts one sixth of the data set.

Put the Performance item's reversed anchors in bold, and consider placing it last rather than fourth so the reversal is not encountered mid-flow.

Scoring

1. Each mark converts to `0-100` in steps of 5 (interval index x 5). If the 0-10 form scale was used, multiply the response by 10 and record the deviation.
2. **Performance is recorded as marked, on its own Perfect-to-Failure scale.** Do not silently invert it. If it is inverted to make the aggregate directionally consistent, state that in the caption, every time.

3. Raw TLX overall score = the unweighted arithmetic mean of the six subscale scores.

Reporting

The six-subscale profile is the primary result. The single aggregate is secondary and reported after it, never instead of it.

With per-task administration the primary artefact is a **six-by-six matrix**: six subscales across six tasks, each cell a mean across participants.

Report as:

- the subscale-by-task matrix as a heatmap or grouped bar chart. This is the headline figure of the evaluation chapter
- per-task aggregate Raw TLX, so the phases can be ranked by total workload
- per-subscale profile collapsed across tasks, for the overall claim
- a table of per-participant, per-task scores in Appendix B so the figures can be recomputed
- median and range, not only means, given the sample size

The matrix is what justifies the per-task cost. If Chapter 9 ends up reporting only the collapsed profile, the six administrations bought nothing and a single end-of-session TLX would have been the right call.

What to look for

The interesting hypothesis for this thesis, stated in advance so that finding it counts and not finding it also counts:

> Apex plausibly **raises Mental Demand** - there is more structure, more vocabulary and more explicit state to hold in mind than in unstructured AI-assisted work - while **lowering Effort and Frustration**, because there is less blank-page work and fewer dead ends.

If that profile appears, it is a precise, defensible claim about *where* the framework moves the practitioner's burden, and it is much more interesting than a low aggregate. If the profile does not appear, report the actual profile and say the hypothesis was not supported. Both outcomes are publishable; only silently reporting the aggregate is not.

Physical Demand will almost certainly be near-floor for a desktop tool. Report it anyway - dropping a subscale because it is uninteresting is a deviation from the instrument.

Per-task predictions, also stated in advance:

Task	Prediction
3, Phase 1 Gherkin	highest Mental Demand - the participant has to express a requirement precisely for the first time
4, Phase 2 design	high Mental Demand, high Effort - most content to review, most decisions to accept or reject
6, Phase 3 packs	moderate throughout - largely a review task
7, Phase 4 QA	low across the board
8, Phase 5 gate	**highest Frustration, lowest Performance** - this task is designed to be refused
9, Export	lowest overall

Task 8's Frustration score is the number to watch. A refusal that the interface explains well should raise Frustration only modestly; a refusal that reads as a bug will spike it. Cross-read that score against UX item A4 and the open question about the same task.

Writing these predictions down before the study is what allows a mismatch to be reported as a finding rather than rationalised afterwards.

Practical note

If the study is run moderated and in person, print the original 20-interval line and use it - it is the instrument, and it costs nothing. The online-form compromise above exists only because the unmoderated arm cannot use paper. Where both arms run, note in Chapter 9 that the two arms used different response scales, and do not pool them into a single mean without saying so.

Apex UI/UX questionnaire - custom instrument

This is not a validated instrument. It was written for this thesis to reach what SUS and Raw TLX structurally cannot see: whether the practitioner trusted the AI output, whether the phase gates read as guardrails or as obstruction, and whether reviewing generated work felt cheaper than writing it.

Consequences that must be honoured in Chapter 9:

- Report **item by item**, descriptively. Diverging bar chart per item.
- **No composite score.** No subscale means, no Cronbach's alpha, no factor analysis. An unvalidated 18-item scale reported as one number is exactly the claim an examiner will take apart.
- The construct groupings below are for organising the discussion, not for computing anything.
- Cite it as an author-designed instrument and say why an off-the-shelf one did not fit.

Administer **last**, after SUS. Roughly 8 minutes.

Scale for all closed items: `1 = Strongly disagree` ... `5 = Strongly agree`, plus an explicit **N/A - did not use this** option. The N/A option matters: several items refer to surfaces a participant may never have opened, and forcing a centre-point answer would manufacture data.

Closed items (EN)

A. Understanding where you are

#	Item
A1	At any moment, I knew which phase of the process I was in.
A2	I understood what the tool expected me to do next.
A3	I could tell the difference between work the tool had finished and work it had not.
A4	When the tool refused to let me continue, I understood why.

A4 is the Task 8 item, the deployment gate that is designed to refuse. It is the highest-value question on this sheet.

B. Working with AI-generated content

#	Item
B1	I could tell which parts of the content were generated by AI and which were mine.
B2	Reviewing the AI's output took less effort than writing it myself would have.
B3	I felt able to reject or change the AI's output when I disagreed with it.
B4	I trusted the generated content enough to use it in a real project.
B5	I could see what information the AI had been given as a basis for its output.

B4 and B2 are the trust-calibration pair. A high B4 with a low B2 says participants trusted output they did not really check, which is a finding about the risk the framework is supposed to control.

B5 targets the Active Context sidebar specifically.

C. Control and reversibility

#	Item
C1	I felt in control of what the tool did on my behalf.
C2	I was confident I could undo or redo a step if I made a mistake.
C3	I understood the consequences of locking a phase before I did it.
C4	Waiting for the AI to finish was acceptable.

C4 exists because Phase 1-4 generation runs 30-60 seconds and the loading feedback has never been usability-tested.

D. Feedback and errors

#	Item
D1	When something went wrong, the tool told me clearly what had happened.
D2	The messages the tool showed me told me what to do next, not only what failed.

D1 and D2 evaluate the error-surfacing work shipped in the 2026-08-04 toast overhaul, which has never been tested with a real user.

E. Fit to real work

#	Item
E1	The artefacts the tool produced are ones my team would actually use.
E2	The process the tool imposes fits the way my team works.
E3	The effort of following the process is worth what it produces.

E2 and E3 evaluate the **framework**, not the tool, and must be reported under the framework's evidence in Chapter 9, not under the instantiation's. Keeping them on this sheet is convenient for the participant but the analysis must split them out. Note this explicitly, or the SUS/TLX separation argued so carefully in `tab:eval_instruments` gets quietly undermined by the author's own questionnaire.

Itens fechados (PT)

Escala: `1 = Discordo totalmente` ... `5 = Concordo totalmente`, mais **N/A - não utilizei**.

A. Perceber onde se está

#	Afirmação
A1	Em qualquer momento, soube em que fase do processo estava.
A2	Percebi o que a ferramenta esperava que eu fizesse a seguir.
A3	Consegui distinguir o trabalho que a ferramenta tinha concluído daquele que não tinha.
A4	Quando a ferramenta me impediu de continuar, percebi porquê.

B. Trabalhar com conteúdo gerado por IA

#	Afirmação
B1	Consegui distinguir que partes do conteúdo foram geradas por IA e quais eram minhas.

#	Afirmação
B2	Rever o resultado da IA exigiu menos esforço do que escrevê-lo eu próprio.
B3	Senti que podia rejeitar ou alterar o resultado da IA quando discordava.
B4	Confiei no conteúdo gerado o suficiente para o usar num projecto real.
B5	Consegui ver que informação tinha sido dada à IA como base para o resultado.

C. Controlo e reversibilidade

#	Afirmação
C1	Senti-me no controlo do que a ferramenta fazia em meu nome.
C2	Senti confiança em conseguir desfazer ou repetir um passo se cometesse um erro.
C3	Percebi as consequências de bloquear uma fase antes de o fazer.
C4	O tempo de espera pela IA foi aceitável.

D. Feedback e erros

#	Afirmação
D1	Quando algo correu mal, a ferramenta disse-me claramente o que tinha acontecido.
D2	As mensagens que a ferramenta mostrou diziam-me o que fazer a seguir, e não apenas o que falhou.

E. Adequação ao trabalho real

#	Afirmação
E1	Os artefactos produzidos pela ferramenta são artefactos que a minha equipa usaria de facto.
E2	O processo que a ferramenta impõe encaixa na forma como a minha equipa trabalha.
E3	O esforço de seguir o processo compensa aquilo que produz.

Open questions

Three, deliberately few. Long open sections at the end of an hour-long session get one-word answers.

EN

1. What was the single most confusing thing about using Apex?
2. Was there a point where you did not trust what the tool had produced? What made you doubt it?
3. If you could change one thing about the interface, what would it be?

PT

1. Qual foi a coisa mais confusa em utilizar o Apex?
2. Houve algum momento em que não confiou naquilo que a ferramenta produziu? O que o levou a duvidar?
3. Se pudesse mudar uma coisa na interface, o que mudaria?

Demographic items

Collected at the **start** of the session, on the consent form, not here. Deferring them to the end costs data when a session overruns.

#	Item
G1	Years of professional software development experience
G2	Current role (developer, tester, PO, designer, manager, student, other)
G3	Frequency of use of AI coding assistants (never / tried it / weekly / daily)
G4	Familiarity with Gherkin or BDD (none / some / regular use)
G5	Familiarity with the project management tool used in the session
G6	Preferred language for the session (EN / PT)

G3 is the important one. Prior exposure to AI assistants will plausibly dominate both trust items and Mental Demand, and it must be reported alongside the results rather than discovered afterwards.

Analysis note

With N around 12 and an unvalidated instrument, the analysis is:

- per-item diverging bar chart, counts not percentages
- median per item
- items where the spread is bimodal are more interesting than items where it is high or low; a split answer usually means the interface works for one mental model and not another, and the interview transcripts will say which
- cross-read A4 against the Task 8 comprehension score from the observer sheet (moderated arm) and against the Task 8 TLX Frustration score (both arms). If participants say they understood the refusal but scored 0 or 1 on explaining it, that gap is a real and reportable finding about confidence exceeding comprehension

Semi-structured interview guides

Two guides, two populations, two purposes. **Both evaluate the framework, not the tool.** SUS, Raw TLX and the UX questionnaire already cover the tool; if these interviews drift into "the button was hard to find", the framework is left with no evidence at all and the whole evaluation collapses onto the instantiation.

Scheduled as a separate session, 30-45 minutes, not immediately after the hour-long task session.

Recorded with explicit consent, transcribed, analysed thematically. Codebook developed from the first two transcripts, then applied to the rest, with codes added as they emerge.

Semi-structured means the questions below are prompts, not a script. Follow what the participant raises. The one hard rule: do not defend the framework. If a participant is wrong about what it does, note the misunderstanding as data and correct it only at the very end of the interview.

Guide 1 - Practitioners

Target: 5-8 people who have used Apex, or who work in the setting where it was demonstrated. Establishes practicality, feasibility and perceived value in real workflows.

Opening (5 min)

1. Describe how your team currently goes from an idea to code in production.
2. Where does AI already appear in that, if at all? Who decided that?
3. What goes wrong most often in that path today?

Purpose: get the baseline in their own words before Apex is mentioned, so that later comparisons are against their real process and not against an idealised one they reconstruct after seeing the framework.

Framework practicality (12 min)

4. The framework splits the lifecycle into six phases with a human decision at the end of each. Which of those decisions would genuinely be made in your team, and which would be rubber-stamped?
5. Which phase would your team resist most? Why that one?
6. The framework requires a specification to exist before code is generated. What does your team do today instead, and what would change?
7. Where would the framework slow you down for no benefit?

Question 7 is the important one and it must be asked directly. A set of interviews in which nobody names a cost is an interview protocol failure, not a validation.

Value and cost (10 min)

8. What would have to be true for your team to adopt this for one project?
9. What would make you abandon it after two sprints?
10. Who in your team would have to own this for it to survive? Does that person exist?
11. If the tool disappeared tomorrow but you kept the framework, would you still use it? Which parts?

Question 11 separates the method from the instantiation directly from the participant's mouth. It is the single strongest piece of interview evidence this thesis can obtain.

Governance and responsibility (8 min)

12. When AI writes code that ships and later fails, who is accountable in your organisation today?
13. Does the framework change that answer? Should it?
14. Does the framework's record of what was decided, and by whom, have value to anyone outside the team? Who?

Close (5 min)

15. What is missing from the framework entirely?
16. Is there anything you expected me to ask and I did not?

Guide 2 - Agile and Scrum experts

Target: 2-3 people with recognised depth in Agile delivery methodology, not necessarily users of Apex. Establishes alignment with, and departure from, established methodology.

Give them the framework document in advance. This interview is about the method on paper; they do not have to have used the tool.

Opening (5 min)

1. How would you describe what this framework is, in your own words?

Purpose: an immediate comprehension check. If an expert cannot restate the framework after reading the document, that is a finding about the framework's clarity, and it belongs in the analytical assessment criterion "clarity".

Alignment and departure (15 min)

2. Where does this align with Scrum, or with Agile principles more broadly?
3. Where does it depart from them? Is the departure justified by what AI changes, or is it a reversion to stage-gated delivery?
4. The framework has explicit gates between phases. Distinguish for me: is that a guardrail or is that a phase gate in the Waterfall sense?
5. Does the framework make the same mistake that heavyweight processes made in the 2000s? What specifically protects it from that, if anything?

Questions 4 and 5 must be asked as bluntly as they are written. "Too linear, this is Waterfall" is already the known criticism from earlier feedback; putting it in front of an expert and recording the answer is far stronger evidence than the author arguing against it.

Roles and artefacts (10 min)

6. The framework treats roles as hats rather than as people. Does that work in a real team, or does it collapse into one person wearing all of them?
7. Which of the required artefacts would you drop? Which would you add?
8. Is the amount of documentation the framework produces proportionate to what it buys?

Adoption (10 min)

9. What kind of team is this framework wrong for?
10. What would a coach have to teach for a team to run this correctly?
11. Would you recommend this to a client? Under what conditions would you refuse?

Close (5 min)

12. If you had to defend this framework to a sceptical engineering manager, what would your strongest argument be? And your weakest point?

Analysis

- Transcribe fully. Do not analyse from notes.
- Codebook from the first two transcripts of each group, then applied across.
- Report themes with representative anonymised quotations, identified as practitioner or expert.
- **Report disconfirming evidence with the same prominence as supporting evidence.** A themes section in which every quotation is favourable will be read, correctly, as selection.
- Cross-read against the UX questionnaire items E1-E3, which are the only framework-level items on that sheet.
- Where practitioners and experts disagree with each other, that disagreement is a finding: it usually marks the gap between what a method claims and what survives contact with a delivery team.

Observer sheet - moderated arm only

Print one per participant. **Arm 2 only** - the unmoderated arm has no observer, which is exactly why arm 2 exists. This sheet produces the observational evidence a questionnaire cannot reach, and partly answers Chapter 9's own admitted threat that "the instruments measure perception rather than delivered software quality".

Task numbering matches `task-script.md`.

```
Participant ID:  P____        Date: __________      Language: EN / PT
Session start: ______    Session end: ______        Remote / In person
Facilitator: ______________
```

Codes

- **Outcome:** `C` completed unaided, `CA` completed with assists, `F` failed or abandoned after three assists.
- **Assist:** any hint given. Write it down **verbatim**. Assists are the most reusable output of the whole study - each one is a specific place the interface failed to communicate.
- **Error:** an action that had to be undone, or a path that could not reach the goal.

Per-task record

```
T1  Sign in and open the project          start ____ end ____   C / CA / F
    Understood the PM tool selector?      Y / N / N/A
    Assists: ______________________________________________
    Errors:  ______________________________________________

T2  Connect to GitHub                     start ____ end ____   C / CA / F
    Looked in Settings first?             Y / N
    Connected state legible?              Y / N
    Assists: ______________________________________________
    Errors:  ______________________________________________

T3  Requirement to locked Gherkin  [TLX]  start ____ end ____   C / CA / F
    Read the clarifying questions?        Y / N / skimmed
    Noticed the assumptions box?          Y / N
```

```
    Hesitated before locking?           Y / N
    Believed the app had hung?          Y / N   at ____ s
    Assists: ______________________________________________
    Errors:  ______________________________________________

T4  Produce and lock the design    [TLX]  start ____ end ____  C / CA / F
    Found the Visual Design group?      Y / N / with assist
    Read the runtime spec?              Y / N / scrolled past
    Edited anything or accepted whole?  edited / accepted
    Assists: ______________________________________________
    Errors:  ______________________________________________

T5  Resume a previous session           start ____ end ____  C / CA / F
    Landed back on the same project?    Y / N
    GitHub still connected?             Y / N       <- a No here is a bug
    Assists: ______________________________________________
    Errors:  ______________________________________________

T6  Implementation tasks          [TLX]  start ____ end ____  C / CA / F
    Named a defensible root task?       Y / N
    Justification, verbatim: ______________________________
    Discovered effort/coverage data?    Y / N
    Read the DAG as a dependency order? Y / N / unclear
    Assists: ______________________________________________

T7  Test plan and QA sign-off     [TLX]  start ____ end ____  C / CA / F
    Understood QA as a human decision?  Y / N / unclear
    Assists: ______________________________________________

T8  The deployment gate           [TLX]  start ____ end ____  C / CA / F
    Read the refusal message aloud?     Y / N
    Looked for an override?             Y / N
    Found the phase-status progression? Y / N

    COMPREHENSION SCORE   2 / 1 / 0
      2 = named the missing QA step and the required order
      1 = knew a process rule blocked them, could not say which
      0 = concluded the tool was broken

    Explanation, verbatim: ________________________________
    _______________________________________________________

T9  Export                        [TLX]  start ____ end ____  C / CA / F
    Export control findable?            Y / N
    Anything downloaded empty/corrupt?  Y / N
    Assists: ______________________________________________
```

Optional tasks, not timed, not counted

```
O1 Phase 6 drift       attempted Y/N   observations: ______________
O2 Traceability        attempted Y/N   observations: ______________
O3 Autopilot           attempted Y/N   observations: ______________
```

Critical incidents

Anything that stopped the participant, confused them for more than 30 seconds, or made them say something unprompted about the tool. One line each, with timestamp and task.

```
____:____   T__   ______________________________________________
____:____   T__   ______________________________________________
____:____   T__   ______________________________________________
____:____   T__   ______________________________________________
____:____   T__   ______________________________________________
____:____   T__   ______________________________________________
```

Verbal answers after T9, verbatim

Ask before handing over the SUS form. Do not discuss the answers.

```
1. What surprised you?
   ____________________________________________________________

2. Was there a moment you did not know what the tool wanted?
   ____________________________________________________________

3. Would you use this on your own project? What would change first?
   ____________________________________________________________
```

Technical failures during the session

Record separately from usability problems. An AI timeout, a 500, a PM API outage or a stale token is not a usability finding, but it contaminates that participant's Raw TLX for the task it happened in and must be reported as a confound against that specific cell of the matrix.

```
____:____   task T__   what failed: ____________________________
```

```
           blocked the task?  Y / N
           recovered by:  participant / facilitator / restart
```

Facilitator debrief, written within 10 minutes

Three sentences, while it is fresh. Worst moment, best moment, and what to change in the protocol before the next session.

```
____________________________________________________________
____________________________________________________________
____________________________________________________________
```

Informed consent, briefing script and demographics

Check with the supervisor whether IST requires formal ethics approval for this study before running the first session. The consent form below is written to be sufficient on its own, but it is not a substitute for an institutional requirement if one exists. This is on the pre-flight checklist.

Which parts apply to which arm

Section	Arm 1, unmoderated	Arm 2, moderated
Consent form	first page of an online form, tick boxes, no signature	printed and signed
Demographics	same online form, immediately after consent	same, on paper
Facilitator briefing script	**not used** - `task-script.md` "Before You Begin" replaces it	read aloud, verbatim
Do-not-do list	not applicable	applies

In arm 1 the consent tick boxes and the demographics questions go on a single short form that the participant completes **before** opening the task script.
Do not bundle them into the SUS or TLX forms: consent has to be given before any data is collected, and demographics answered at the end get skipped by anyone whose session overran.

Arm 1 consent wording needs two small changes from the printed text below: replace "Signature" with a required tick box reading "I confirm I have read and agree to the above", and drop the screen-recording item unless the unmoderated sessions are actually being recorded, which by default they are not.

Informed consent form (EN)

Study: Evaluation of Apex, a tool supporting a process framework for human-AI collaboration in software development.
Researcher: Tomás dos Santos Taborda, MEIC-T, Instituto Superior Técnico, Universidade de Lisboa.
Supervisors: Prof. Miguel Mira da Silva, Hugo de Sousa.
Contact: tomassantostaborda@tecnico.ulisboa.pt

What this involves. You will use a software tool for about 60 minutes to complete a set of predefined tasks, thinking aloud as you go. Afterwards you will fill in three short questionnaires, taking about 20 minutes. You may optionally be invited to a separate interview of 30 to 45 minutes.

What is being tested. The tool and the process behind it are being tested, not you. There is no way for you to perform badly. Difficulties you have are findings about the tool and are the most useful thing you can give us.

What is recorded. Your answers to the questionnaires, notes taken by the observer, the times you take on each task, and, with your separate permission below, a screen and audio recording of the session.

Anonymity. You are identified only by a code such as P3. Your name, your employer and any project details you mention are not recorded in the results. Quotations used in the dissertation are anonymised, and any detail that could identify you or your organisation is removed or generalised.

Data handling. Recordings are stored encrypted, are accessible only to the researcher, and are deleted after transcription and no later than the conclusion of the dissertation. Anonymised questionnaire responses and transcripts are kept so that reported results can be independently recomputed, and may be reproduced in an appendix of the dissertation in anonymised form.

Voluntary participation. Participation is voluntary. You may take a break at any time, skip any task or question, or stop and withdraw entirely without giving a reason and without any consequence. If you withdraw, tell the researcher whether your data so far should be deleted or kept.

Risks. There are no anticipated risks beyond mild frustration with software.

```
I have read and understood the above, and I agree to take part.

[ ] I consent to participate.
[ ] I consent to the session being screen and audio recorded.
[ ] I consent to anonymised quotations being used in the dissertation.
[ ] I agree to be contacted about a follow-up interview.

Name (block capitals): ______________________________
Signature: __________________________  Date: ____________
```

Declaração de consentimento informado (PT)

Estudo: Avaliação do Apex, uma ferramenta de apoio a uma framework de processo para colaboração humano-IA no desenvolvimento de software.
Investigador: Tomás dos Santos Taborda, MEIC-T, Instituto Superior Técnico, Universidade de Lisboa.
Orientadores: Prof. Miguel Mira da Silva, Hugo de Sousa.
Contacto: tomassantostaborda@tecnico.ulisboa.pt

Em que consiste. Vai utilizar uma ferramenta de software durante cerca de 60 minutos para realizar um conjunto de tarefas predefinidas, verbalizando o seu raciocínio. Depois preencherá três questionários curtos, cerca de 20 minutos. Poderá ainda ser convidado para uma entrevista separada de 30 a 45 minutos.

O que está a ser testado. Está a ser testada a ferramenta e o processo subjacente, não o participante. Não é possível ter um mau desempenho. As dificuldades que sentir são resultados sobre a ferramenta e são o contributo mais útil que nos pode dar.

O que é registado. As suas respostas aos questionários, notas do observador, os tempos de cada tarefa e, mediante autorização separada abaixo, gravação de ecrã e áudio da sessão.

Anonimato. É identificado apenas por um código, por exemplo P3. O seu nome, a sua entidade empregadora e quaisquer detalhes de projectos que mencione não são registados nos resultados. As citações utilizadas na dissertação são anonimizadas e qualquer detalhe que o possa identificar, a si ou à sua organização, é removido ou generalizado.

Tratamento dos dados. As gravações são guardadas cifradas, acessíveis apenas ao investigador, e são eliminadas após transcrição e nunca depois da conclusão da dissertação. As respostas anonimizadas e as transcrições são conservadas para que os resultados possam ser recalculados de forma independente, e podem ser reproduzidas de forma anonimizada num anexo da dissertação.

Participação voluntária. A participação é voluntária. Pode fazer uma pausa a qualquer momento, saltar qualquer tarefa ou pergunta, ou desistir por completo sem indicar qualquer motivo e sem qualquer consequência. Se desistir, indique ao investigador se os dados já recolhidos devem ser eliminados ou mantidos.

Riscos. Não são antecipados riscos para além de alguma frustração com software.

```
Li e compreendi o acima descrito e aceito participar.

[ ] Consinto em participar.
[ ] Consinto na gravação de ecrã e áudio da sessão.
[ ] Consinto na utilização de citações anonimizadas na dissertação.
[ ] Aceito ser contactado para uma entrevista de seguimento.

Nome (maiúsculas): ______________________________
Assinatura: __________________________  Data: ____________
```

Demographics form

Filled in at the start, immediately after consent. Attach to the observer sheet.

```
Participant ID: P____

G1  Years of professional software development experience
    [ ] under 1   [ ] 1-3   [ ] 4-7   [ ] 8-15   [ ] over 15   [ ] student

G2  Current role
    [ ] developer  [ ] tester/QA  [ ] product owner  [ ] designer
    [ ] engineering manager  [ ] student  [ ] other: __________

G3  How often do you use AI coding assistants?
    [ ] never  [ ] tried once or twice  [ ] monthly  [ ] weekly  [ ] daily
```

```
G4  Familiarity with Gherkin or BDD
    [ ] none   [ ] have seen it   [ ] use it regularly

G5  Familiarity with the project management tool used today
    [ ] none   [ ] some   [ ] use it regularly

G6  Preferred language for this session
    [ ] English   [ ] Português
```

G3 is expected to be the strongest confound in the results and must be reported alongside them.

Facilitator briefing script

Read this out. Do not improvise it. Improvising the briefing is how a facilitator accidentally teaches the participant the interface.

> Thank you for taking part. This will take about 80 minutes.
>
> I am going to ask you to complete six tasks in a tool called Apex. I want to be clear that we are testing the tool, not you. If something is confusing or you cannot find something, that is exactly what I need to know. There is no way for you to get this wrong.
>
> Please think aloud as you work. Say what you are looking at, what you expect to happen, and what you are trying to do. If you go quiet I will ask you what you are thinking, and that is not a hint, it is just a reminder.
>
> I will read each task out and I will not help you unless you are stuck for a while. If I do help, that is fine and it is useful information for me. If you want to give up on a task, say so and we will move on.
>
> The tool uses AI to generate content, and some steps take up to a minute. That is normal.
>
> You can stop, take a break, or leave at any point without giving a reason.
>
> Do you have any questions before we begin?

Then a **five-minute orientation**, identical for every participant, not scored: show where the phases are, where the sidebar is, and where settings are. Nothing task-specific. Write down what you showed, and show exactly that every time.

Facilitator do-not-do list

- Do not point at the screen.
- Do not say "you could try...".
- Do not react to a mistake, verbally or facially.
- Do not explain what the tool was supposed to do.
- Do not defend a design decision, even if the participant asks directly. Say "I will explain everything at the end" and note the question.
- Do not skip the observer sheet during a busy session. Notes reconstructed afterwards are unreliable, and unreliable notes are worse than none.